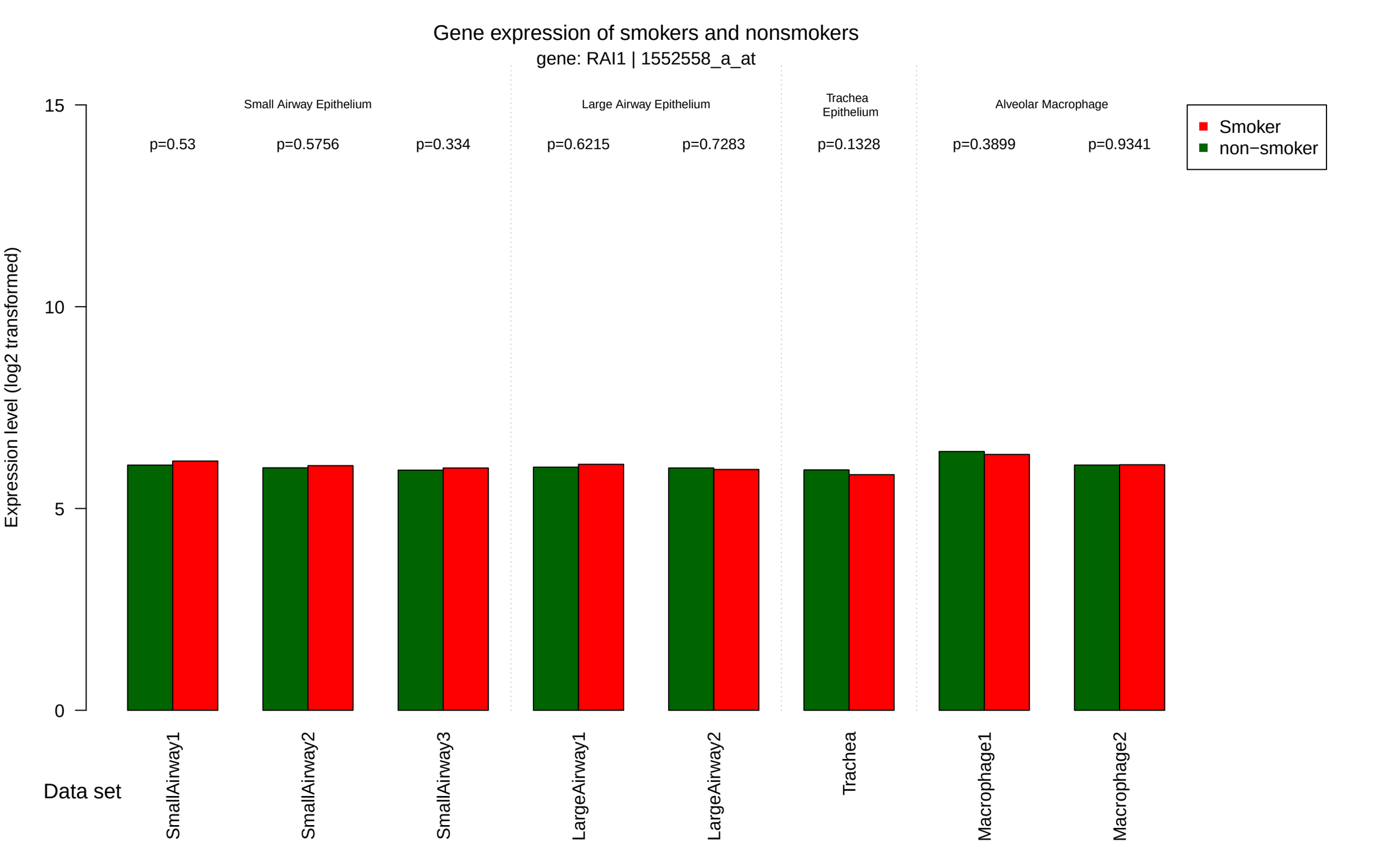
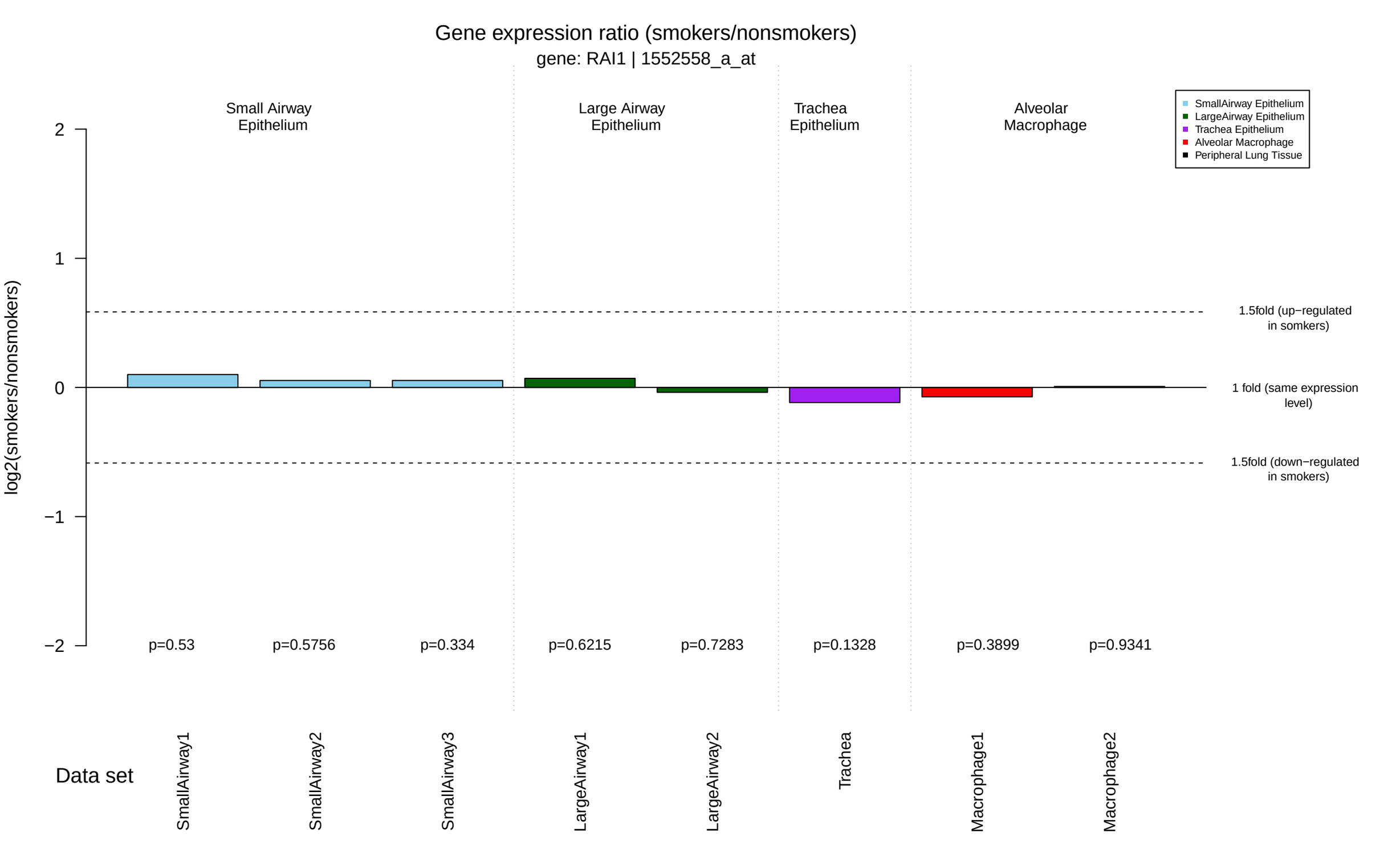
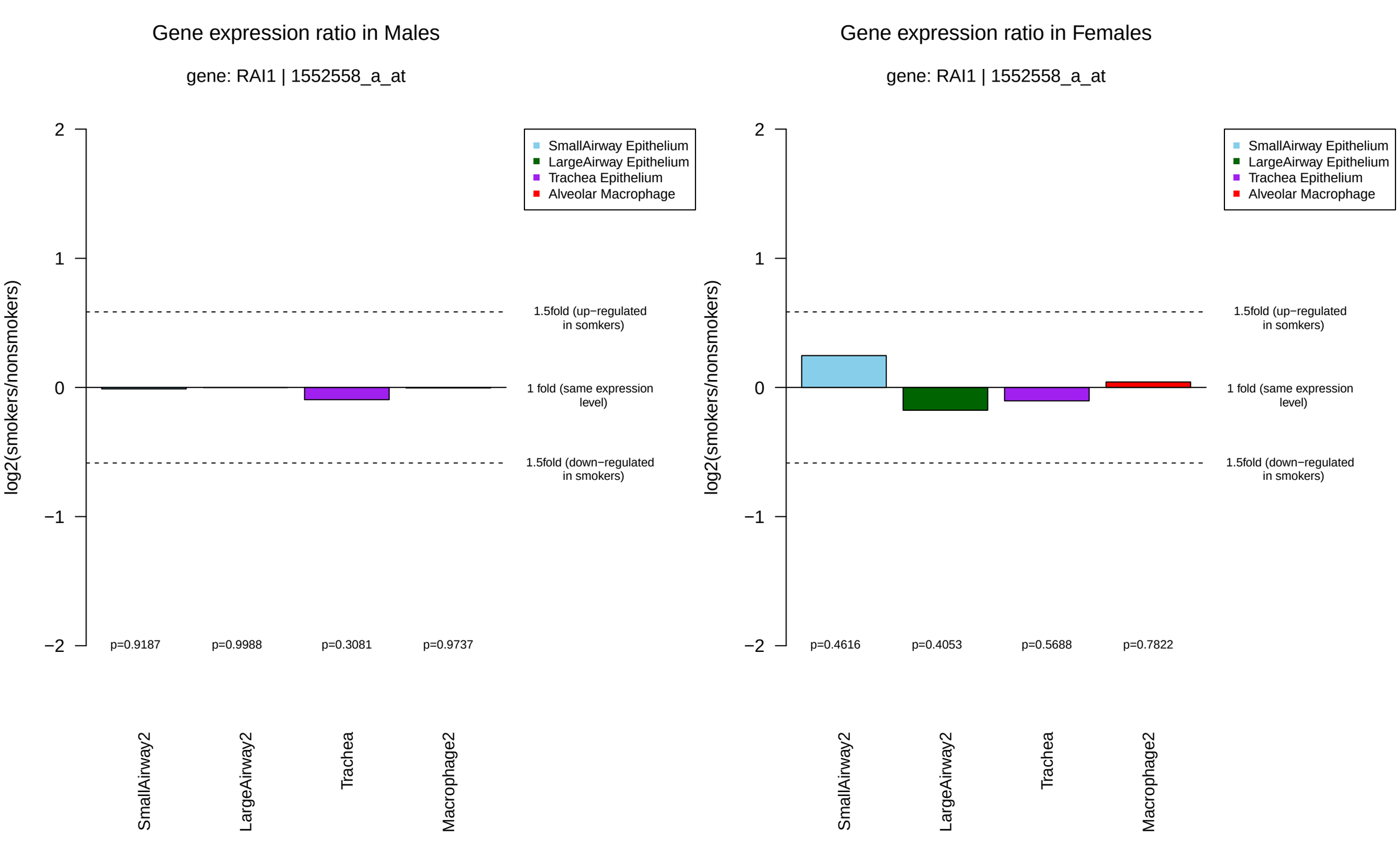
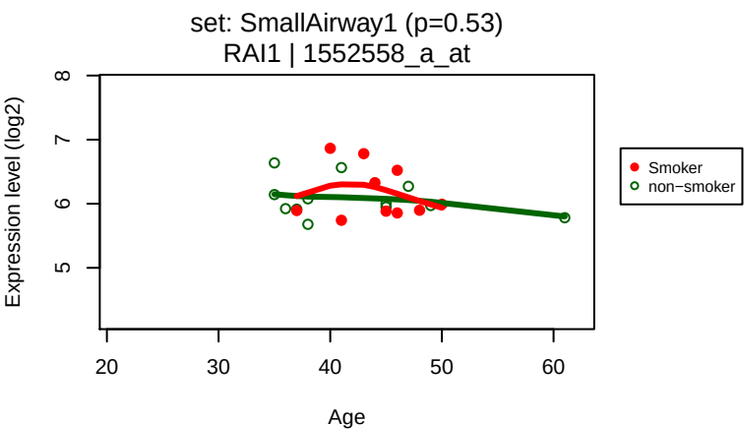
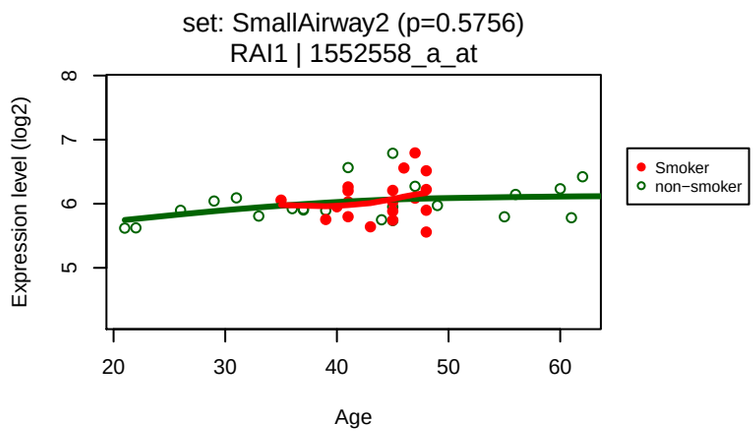
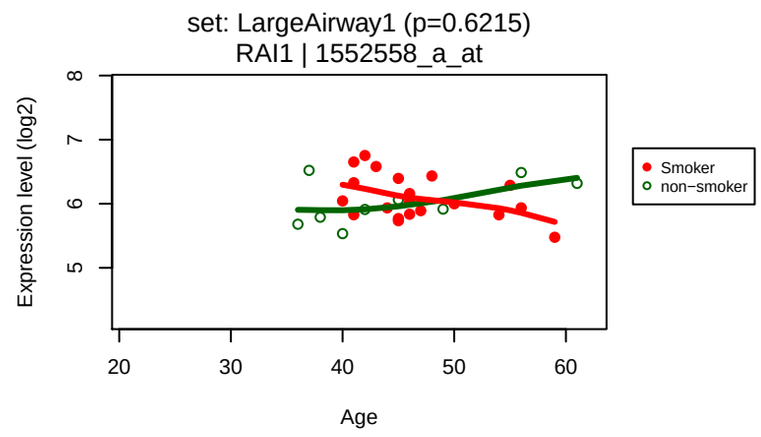

Gene expression of smokers and nonsmokers
gene: RAI1 | 1552558_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.53
p=0.5756
p=0.334
p=0.6215
p=0.7283
p=0.1328
p=0.3899
p=0.9341
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

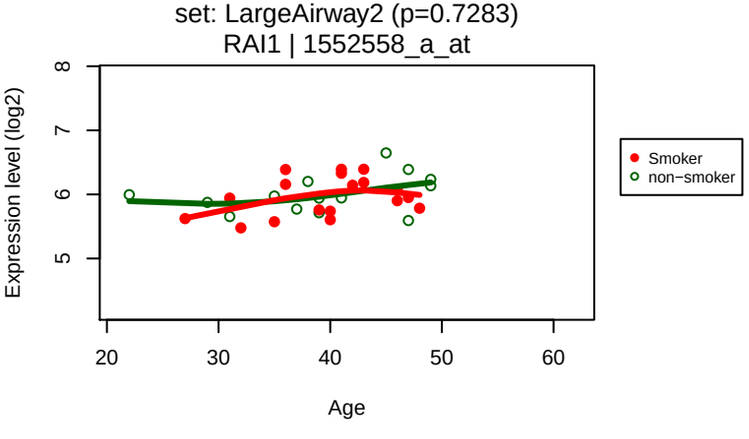

Gene expression ratio (smokers/nonsmokers)
gene: RAI1 | 1552558_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.53
p=0.5756
p=0.334
p=0.6215
p=0.7283
p=0.1328
p=0.3899
p=0.9341
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

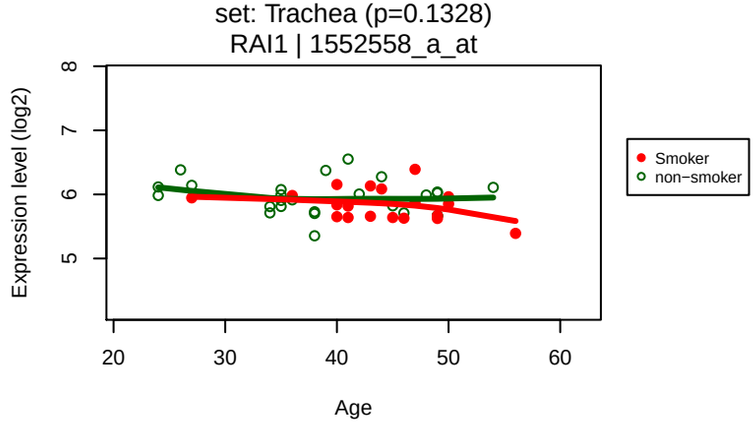

Gene expression ratio in Males
gene: RAI1 | 1552558_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9187
p=0.9988
p=0.3081
p=0.9737
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAI1 | 1552558_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4616
p=0.4053
p=0.5688
p=0.7822
SmallAirway2
LargeAirway2
Trachea
Macrophage2

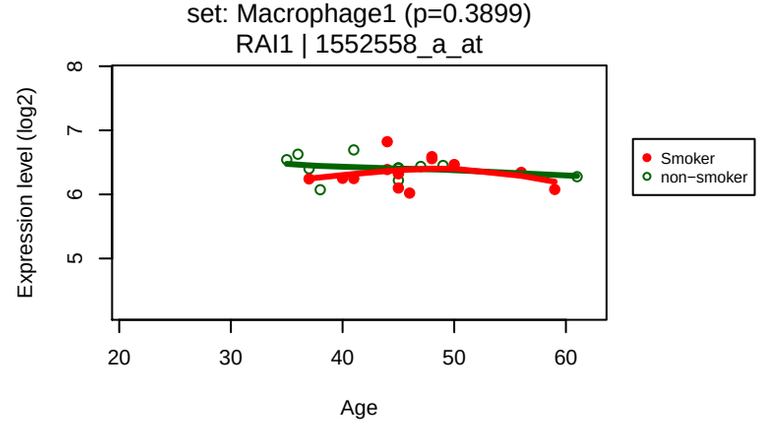

set: SmallAirway1 (p=0.53)
RAI1 | 1552558_a_at
Expression level (log2)
Age
Smoker
non−smoker

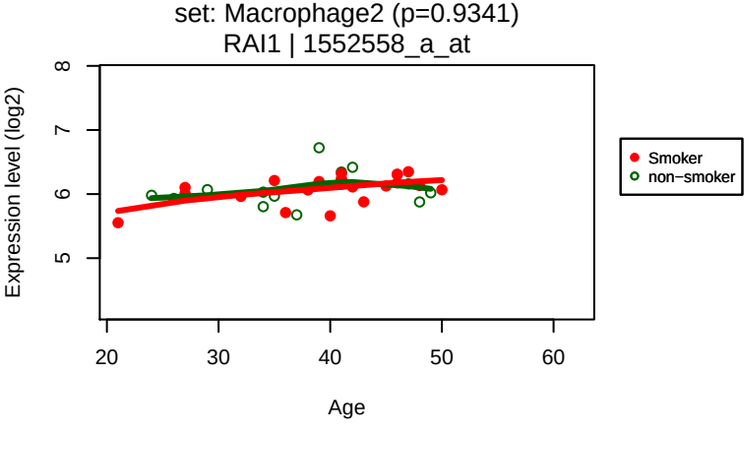

set: SmallAirway2 (p=0.5756)
RAI1 | 1552558_a_at
Expression level (log2)
Age
Smoker
non−smoker

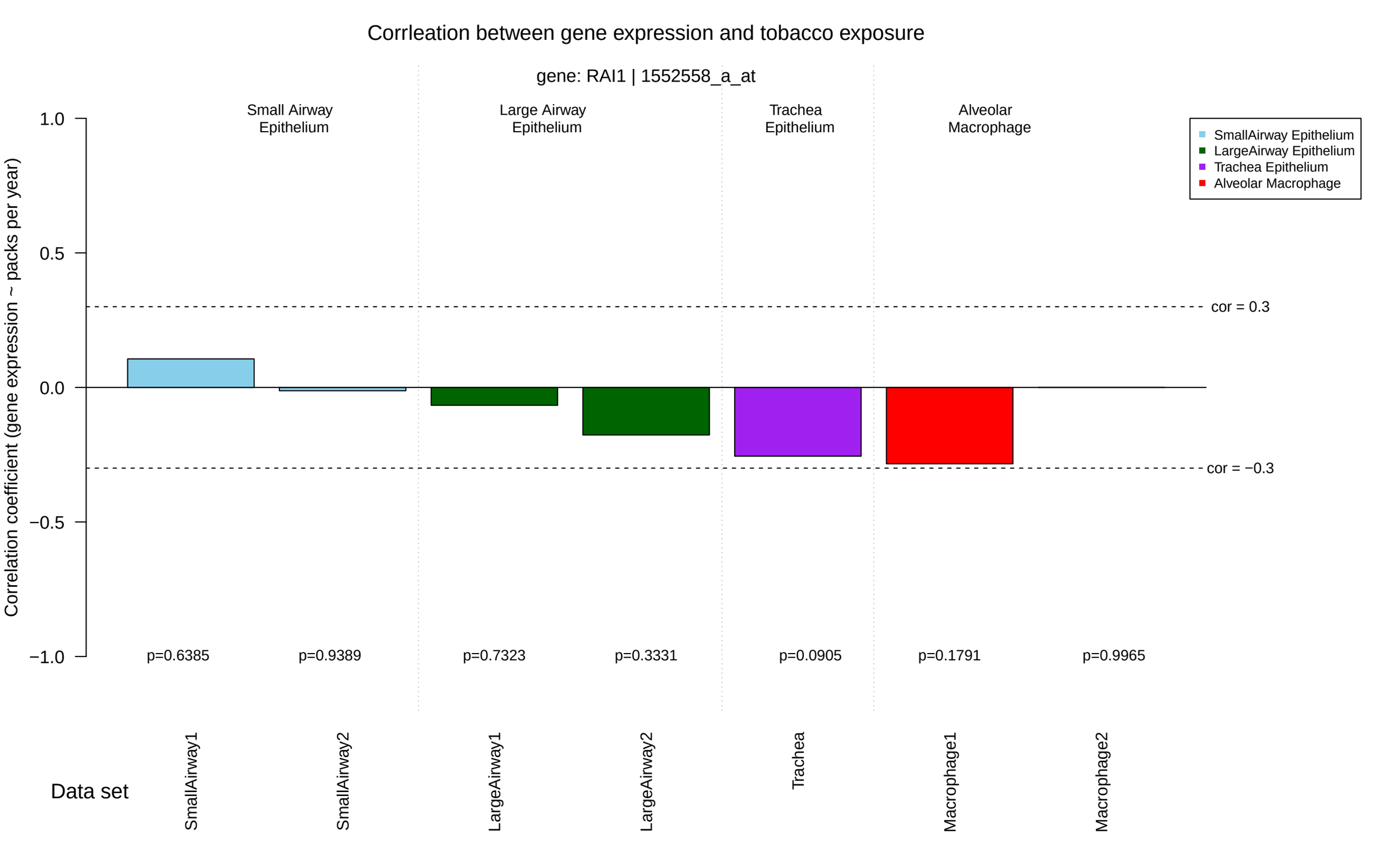

set: LargeAirway1 (p=0.6215)
RAI1 | 1552558_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7283)
RAI1 | 1552558_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1328)
RAI1 | 1552558_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3899)
RAI1 | 1552558_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9341)
RAI1 | 1552558_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAI1 | 1552558_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6385
p=0.9389
p=0.7323
p=0.3331
p=0.0905
p=0.1791
p=0.9965
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2